



بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

University of Sinnar

Faculty of Nursing Sciences



**Assessment of knowledge and Practice of Nurses and
Midwives Regarding Immediate Post Cesarean Care At
Obstetrical and Gynecological Sinnar Teaching Hospital**

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The verse

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

قال تعالى:

(وَوَصَّيْنَا الْإِنْسَانَ بِوَالِدَيْهِ إِحْسَانًا حَمَلَتْهُ أُمُّهُ كُرْهًا وَوَضَعَتْهُ كُرْهًا وَحَمَلُهُ
وَفَصْلُهُ ثَلَاثُونَ شَهْرًا حَتَّىٰ إِذَا بَلَغَ أَشُدَّهُ وَبَلَغَ أَرْبَعِينَ سَنَةً قَالَ رَبِّ أَوْزِعْنِي
أَنْ أَشْكُرَ نِعْمَتَكَ الَّتِي أَنْعَمْتَ عَلَيَّ وَعَلَىٰ وَالِدَيَّ وَأَنْ أَعْمَلَ صَالِحًا تَرْضَاهُ
وَأَصْلِحْ لِي فِي ذُرِّيَّتِي إِنِّي تُبْتُ إِلَيْكَ وَإِنِّي مِنَ الْمُسْلِمِينَ)

صدق الله العظيم

سورة الأحقاف الآية (15)

Dedication

We dedicate this work with deepest gratitude and appreciation to our beloved parents, whose unconditional love, endless sacrifices, unwavering support, and sincere prayers have been the foundation of our success throughout our academic journey

We are equally grateful to our families for their patience, understanding, encouragement, and constant belief in our abilities during every stage of this work.

We extend our heartfelt appreciation to our supervisors, teachers, and mentors for their invaluable guidance, knowledge, wisdom, and continuous inspiration, which greatly enriched this research.

Special thanks are due to our friends and colleagues for their support, cooperation, motivation, and for sharing with us the challenges and achievements along this journey.

Finally, we dedicate this work to everyone who contributed, directly or indirectly, to the successful completion of this study.

Researchers

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Researchers

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List of Abbreviation

BP	Blood Pressure
CS	Cesarean Section
DVT	Deep Vein Thrombosis
HR	Heart Rate
IPC	Immediate Post Cesarean Care

Abstract

Background: Immediate post–cesarean section care is a critical component of maternal health services aimed at preventing postoperative complications and ensuring safe recovery. This study assessed the knowledge and practices of nurses and midwives regarding immediate post–cesarean section care at Sinnar Teaching Hospital.

Methods: A descriptive cross-sectional hospital-based study was conducted among nurses and midwives working in obstetric and gynecological units. Data were collected using a structured questionnaire assessing participants' knowledge and clinical practices. Data were analyzed using descriptive and inferential statistics, and results were presented in tables, percentages, chi-square test, and correlation analysis.

Results: The majority of participants were nurses (73.3%), while midwives accounted for 26.7%. Most respondents held diploma qualifications (66.7%), and 56.7% had not received formal training in post-cesarean care. Overall, participants demonstrated a moderate level of knowledge (67%) and generally acceptable practice in routine postoperative care, including monitoring vital signs (93.3%), urine output (93.3%), and wound assessment (90%). However, knowledge and practice gaps were identified in areas such as pharmacological side effects, pain assessment, early breastfeeding initiation, patient education, and deep vein thrombosis prevention (53.3%).

Conclusion: The study concluded that nurses and midwives at Sinnar Teaching Hospital have a moderate level of knowledge regarding immediate post-cesarean section care. Despite acceptable performance in basic postoperative care, deficiencies in specialized knowledge and patient-centered interventions were evident.

المستخلص

المقدمة: تُعد الرعاية الفورية بعد الولادة القيصرية جزءاً أساسياً من خدمات صحة الأم، وتهدف إلى منع المضاعفات بعد الجراحة وضمان تعافٍ آمن. هدفت هذه الدراسة إلى تقييم معرفة وممارسات الممرضين والقابلات فيما يتعلق بالرعاية الفورية بعد الولادة القيصرية في مستشفى سنار التعليمي.

المنهجية: أجريت دراسة وصفية مقطعية داخل المستشفى بين الممرضين والقابلات العاملين في أقسام النساء والتوليد. تم جمع البيانات باستخدام استبيان منظم لقياس المعرفة والممارسات السريرية للمشاركين. تم تحليل البيانات باستخدام الإحصاء الوصفي والاستدلالي، وعُرضت النتائج في جداول ونسب مئوية واختبار كاي تربيع وتحليل الارتباط.

النتائج: أظهرت النتائج أن معظم المشاركين كانوا من الممرضين بنسبة 73.3%، بينما شكلت القابلات 26.7%. وكان أغلب المشاركين يحملون مؤهل الدبلوم بنسبة 66.7%، كما أن 56.7% منهم لم يتلقوا تدريباً رسمياً في رعاية ما بعد الولادة القيصرية. بشكل عام، أظهر المشاركون مستوى معرفة متوسطاً 67% وممارسات مقبولة في الرعاية الروتينية بعد العملية، بما في ذلك مراقبة العلامات الحيوية 93.3%، وإدراج البول 93.3%، وتقييم الجرح 90% ومع ذلك، وُجدت فجوات في المعرفة والممارسة في مجالات مثل الآثار الجانبية للأدوية، تقييم الألم، البدء المبكر للرضاعة الطبيعية، التنقيف الصحي، والوقاية من الجلطات الوريدية العميقة 53.3%.

الخلاصة: خلصت الدراسة إلى أن الممرضين والقابلات في مستشفى سنار التعليمي لديهم مستوى معرفة متوسط حول الرعاية الفورية بعد الولادة القيصرية. وعلى الرغم من الأداء المقبول في الرعاية الأساسية بعد الجراحة، إلا أن هناك قصوراً في المعرفة المتخصصة والتدخلات المتمحورة حول المريض.

1.1 Introduction

1.1 Background

Cesarean section is one of the most commonly performed obstetric surgical procedures worldwide and plays a vital role in reducing maternal and neonatal morbidity and mortality when complications arise during childbirth⁽¹⁾.

The immediate post-cesarean period is a critical phase⁽²⁾, as women are at increased risk of complications such as hemorrhage, infection, pain, thromboembolism, and delayed initiation of breastfeeding⁽³⁾.

Effective and timely post-cesarean care is therefore essential to promote maternal recovery, ensure neonatal well-being, and prevent adverse outcomes.

Nurses and midwives are key healthcare professionals responsible for providing immediate post-cesarean care⁽⁴⁾. Their roles include close monitoring of vital signs, assessment of surgical wounds, pain management, prevention of complications, support for early mother-infant bonding, and education of mothers regarding self-care and newborn care⁽⁵⁾.

The quality of care delivered during this period largely depends on the nurses' and midwives' knowledge, clinical skills, and adherence to evidence-based practices⁽⁶⁾.

1.1.2 Problem Statement

According to the World Health Organization (WHO), the ideal rate of cesarean deliveries should range between 10% and 15%⁽⁷⁾. However, global statistics have shown a significant increase, with the number of cesarean births rising from approximately 16 million in 2000 to nearly 29.7 million in 2020⁽⁸⁾.

This rise raises concerns about potential overuse and associated complications such as hemorrhage, infection, thrombosis, or even maternal and fetal death ⁽⁷⁾

Therefore, assessing the knowledge and practices of healthcare providers regarding post-cesarean care is essential to ensure safe recovery and improve maternal outcomes ⁽⁸⁾.

1.1.3 Justification

Midwives play a comprehensive role in caring for the mother after a cesarean section, starting from monitoring the condition and quantity of the placenta to providing guidance and planning for future deliveries ⁽⁴⁾

Nurses manage the care of the mother from the moment she is admitted to the ward until she is discharged in good health with her baby ⁽⁵⁾

Based on this vital role, any deficiency in the knowledge or practice of midwives and nurses regarding immediate post-cesarean care can lead to serious health complications that negatively affect the mother's recovery and safety ⁽⁶⁾

Therefore, it is important to assess their knowledge and practices to ensure the provision of safe and effective care ⁽⁸⁾

1.1.4 Objectives

1.1.4.1 General Objective

Assessment of the knowledge and practices of nurses and midwives regarding immediate post-cesarean care at Sinnar Teaching Hospital, Obstetrics and Gynecology.

1.1.4.2 Specific Objectives

Identify the level of knowledge among nurses and midwives about essential post-cesarean care procedures.

Evaluate the extent of adherence to proper clinical practices in immediate post-cesarean care.

1.2 Literature Review

1.2.1 Overview of Cesarean Section

Cesarean section (CS) is defined by the World Health Organization as a surgical procedure used to deliver a baby through incisions in the abdomen and uterus ⁽¹⁾

It is a critical intervention for reducing maternal and neonatal mortality when medically indicated Over the past decades ⁽²⁾.The global rate of cesarean section has increased significantly, raising concerns about both overuse and underuse ⁽³⁾

The WHO emphasizes that unnecessary cesarean sections may expose mothers and newborns to avoidable complications, while lack of access to the procedure when needed can increase mortality.

1.2.2 Immediate Post-Cesarean Care

Immediate post-cesarean care is a crucial phase that focuses on ensuring maternal stability and preventing complications ⁽²⁾. According to the World Health Organization, essential components of postnatal care include monitoring vital signs, assessing uterine contraction, observing vaginal bleeding, and evaluating the level of consciousness ⁽¹⁾.

Early identification of complications such as postpartum hemorrhage, infection, and thromboembolism is vital during this period (2).

Effective postoperative care also includes pain management, wound care, and early mobilization, all of which contribute to faster recovery and reduced hospital stay ⁽⁵⁾.

1.2.3 Role of Nurses and Midwives

Nurses and midwives play a fundamental role in delivering immediate post-cesarean care ⁽⁴⁾.

The World Health Organization highlights that they are central to providing safe, effective, and patient-centered care.

Their responsibility include

Continuous monitoring of maternal condition, Early detection and reporting of complications, Administration of medications and pain management, Providing wound care and infection prevention, Supporting early initiation of breastfeeding ⁽⁵⁾. Their role is particularly important in low-resource settings, where they may be the primary healthcare providers responsible for maternal care ⁽⁸⁾.

1.2.4 Breastfeeding and Maternal Support

The World Health Organization recommends early initiation of breastfeeding within the first hour after birth, including after cesarean section, when possible ⁽⁴⁾.

Nurses and midwives are responsible for assisting mothers in positioning and attachment, as well as overcoming challenges related to postoperative pain ⁽⁴⁾.

In addition to physical care, psychological support is essential. Women undergoing cesarean section may experience anxiety, fear, or delayed bonding with their newborn.

Support from healthcare providers plays a key role in improving maternal satisfaction and emotional well-being ⁽⁶⁾.

1.2.5 Knowledge and Practices of Nurses and Midwives

The quality of post-cesarean care is highly dependent on the knowledge and practices of nurses and midwives ⁽⁶⁾

The World Health Organization emphasizes the importance of education, training, and adherence to evidence-based guidelines to ensure optimal outcomes ⁽⁵⁾

Studies have shown that gaps in knowledge or inadequate clinical practice can lead to increased risk of complications, including infection and delayed recovery ²).

Therefore, continuous professional development and training programs are essential to improve the competence of healthcare providers ²).

1.2.6 Challenges in Post-Cesarean Care

Despite the availability of guidelines, several challenges affect the quality of care, particularly in developing countries ⁽⁴⁾.

These include shortage of trained staff, lack of resources, high workload, and limited access to updated clinical guidelines ⁽⁸⁾.

The World Health Organization stresses the need for strengthening health systems and investing in the nursing and midwifery workforce to address these challenges ⁽

¹)

2. Methodology

2.1 Research Design

This is a descriptive cross-sectional hospital-based study conducted to assess the knowledge and practice of nurses and midwives regarding immediate post-cesarean care at Sinnar Obstetrics and Gynecology Teaching Hospital.

2.2 Study Area

The study was conducted at Sinnar Obstetrics and Gynecology Teaching Hospital, located in the eastern part of Sinnar City, Sudan. The hospital became an independent specialized hospital in 2014 after previously being part of Sinnar Teaching Hospital. It provides specialized maternal and women's healthcare services and receives referrals from different areas of Sinnar State and neighboring regions.

The hospital includes several departments such as the Bleeding Ward, Intensive Care Unit (ICU), Cesarean Section Ward, Cesarean Section Operating Room, Postnatal Ward, and Neonatal Unit. The study focused on nurses and midwives working in departments related to post-cesarean care.

2.3 Study Population(participants)

All nurses and midwives working at Sinnar Obstetrics and Gynecology Teaching Hospital during the study period.

2.4 Inclusion Criteria

Nurses and midwives currently working in the Sinnar Maternity Teaching Nurses and midwives with more than six months of experience in post-cesarean care.

Those who agreed to participate in the study.

2.5 Exclusion Criteria

Nurses and midwives with less than six months of experience in post-cesarean care.

Nurses and midwives who were absent during the data collection period.

Those who refused to participate in the study.

2.6 Sampling

A non-probability convenience sampling technique was used in this study. All available nurses and midwives working in the Sinnar Maternity Teaching Hospital who met the inclusion criteria during the data collection period were invited to respondents .

The sample size is determined according to the statistical formula:

$$n = \frac{Z^2 pq}{d^2}$$

Where:

Z = index of confidence level which is (z= 1.96 at 95% confidence interval)

p = Prevalence of phenomena in this study is set to be (0.02 or 2% of patients who admitted to Sinnar Teaching Hospital)

q = 1- p

d = error margin (for 95% confidence, d = 0.05 or 5%)

n = sample size

$$n = \frac{1.96^2 0.02(1 - 0.02)}{0.05^2} = 30.11 = 30 \text{ participants}$$

2.7 Variables (Independent and Dependent)

Independent variables: age, gender, profession, educational qualification, and years of experience.

Dependent variables: knowledge and practice of nurses and midwives regarding immediate post-cesarean care.

2.8 Data Collection Technique and Tools

Tool: A structured questionnaire and observational checklist were used to collect data on demographic characteristics, knowledge, and practice regarding immediate post-cesarean care.

Technique: Face-to-face interviews were conducted with participants using the questionnaire, while practice was assessed through an observational checklist.

The questionnaire included demographic data and close-ended questions covering key areas such as monitoring vital signs, pain management, monitoring vaginal bleeding, uterine assessment, signs of wound infection, prevention of leg thrombosis, and breastfeeding support.

2.9 Data Analysis

Data were entered, cleaned, and analyzed using the Statistical Package for the Social Sciences (SPSS) version 26.

Knowledge items were scored as (1) for correct answers and (0) for incorrect answers. Practice items were scored as (1) for "Done" and (0) for "Not Done."

Descriptive statistics including frequencies, percentages, means, and standard deviations were used to summarize the data.

2.10 Ethical Consideration

Ethical approval was obtained from the University of Sinnar, Faculty of Nursing Science, and Sinnar Maternity Teaching Hospital. Verbal informed consent was obtained from all participants after explaining the study objectives.

Confidentiality was maintained, and participants had the right to withdraw at any stage of the study.

3.Result

Table (3.1): Socio-demographic Characteristics of the Participants

Table 1: Distribution of participants by socio-demographic data (N=30)

Variable	Category	Frequency (f)	Percentage (%)
Age	<25 years	8	26.7%
	25 - 35 years	15	50.0%
	35 - 40 years	5	16.6%
	>40 years	2	6.7%
Profession	Nurse	22	73.3%
	Midwife	8	26.7%
Qualification	Diploma	20	66.7%
	Bachelor (B.Sc)	9	30.0%
	Master (M.Sc)	1	3.3%
Experience	<1 year	4	13.3%
	1 - 3 years	9	30.0%
	3 - 6 years	11	36.7%
	> 6 years	6	20.0%
Training	Trained	13	43.3%
	Not Trained	17	56.7%

Table(3.2): Assessment of Nurses' and Midwives' Knowledge

Table 2: Participants' responses to knowledge items regarding post-CS care (N=30)

NO	Knowledge Item [cite: 6-17]	Correct N(%)	Incorrect N(%)
1	Vital signs monitoring every 15 min	24 (80.0%)	6 (20.0%)
2	Monitoring during first 2 hours	26 (86.7%)	4 (13.3%)
3	Pain assessment using verbal scale	19 (63.3%)	11 (36.7%)
4	Morphine & Anesthesia side effects	15 (50.0%)	15 (50.0%)
5	Tachycardia as bleeding indicator	28 (93.3%)	2 (6.7%)
6	Signs of wound infection	25 (83.3%)	5 (16.7%)
7	Repeated check of surgical dressing	27 (90.0%)	3(10.0%)
8	Signs of thrombosis	22 (73.3%)	8 (26.7%)
9	Importance of early mobilization	23 (76.7%)	7 (23.3%)
10	Uterine massage technique	21 (70.0%)	9 (30.0%)
11	Barriers to improving care	24 (80.0%)	6 (20.0%)
12	Monitoring urine output	28 (93.3%)	2 (6.7%)
13	Early breastfeeding initiation	20 (66.7%)	10 (33.3%)
14	Observation of lochia	26 (86.7%)	4(13.3%)

Table (3.3): Assessment of Practice (Observational Checklist)

Table 3: Practice levels of participants during post-cesarean care (N=30)

NO	Observed Practice	Done N (%)	Not Done N (%)
1	V/S check upon arrival from OR	28)93.3%(	2 (6.7%)
2	V/S every 15 min (1st hour)	22 (73.3%)	8 (26.7%)
3	Vaginal bleeding check	24 (80.0%)	6 (20.0%)
4	Uterine fundus palpation	18 (60.0%)	12 (40.0%)
5	Incision site assessment	27 (90.0%)	3(10.0%)
6	Level of consciousness assessment	25 (83.3%)	5 (16.7%)
7	Encouraging leg movement	16 (53.3%)	14 (46.7%)
8	Pain management/Analgesia	21 (70.0%)	9 (30.0%)
9	Breastfeeding assistance	23 (76.7%)	11 (36.7%)
10	Urine output monitoring	26 (86.7%)	4 (13.3%)
11	Documentation in patient file	23 (76.7%)	7 (23.3%)
12	Health education on danger signs	15 (50.0%)	15 (50.0%)

Table (3.4): Bivariate Analysis (Inferential Statistics)

Table 4: Relationship between Training and Overall Practice Level

Training Status	Good Practice	Poor Practice	Total	\chi ²	P-value
Trained	11	2	13	7.42	0.006
Not Trained	6	11	17		

Table (3.5): Correlation Analysis

Table 5: Correlation between Total Knowledge and Total Practice Scores

Variables	Mean Score	Std. Deviation	Pearson Correlation (r)	Sig. (2-tailed)
Total Knowledge	10.9	2.1	0.68	0.001
Total Practice	8.8	2.4		

4.1 Discussion

The present study assessed the knowledge and practices of nurses and midwives regarding immediate post-cesarean section care at Sinnar Teaching Hospital. The findings revealed that the majority of participants were nurses (73.3%), while midwives represented only 26.7%. Most participants held diploma qualifications (66.7%), and half of them were between 25–35 years old. In addition, more than half of the participants (56.7%) had not received formal training related to post-cesarean care. This finding may indicate limited opportunities for continuous professional education and in-service training within the hospital setting. Lack of training could negatively influence both knowledge and practical performance among healthcare providers.

Regarding knowledge assessment, the study demonstrated that participants had good knowledge in several essential aspects of immediate post-cesarean care. Most participants correctly identified tachycardia as an indicator of internal bleeding (93.3%) and recognized the importance of monitoring urine output (93.3%). High knowledge levels were also observed in wound infection signs, repeated surgical dressing assessment, and monitoring during the first two postoperative hours. These findings suggest that nurses and midwives are familiar with routine postoperative observations and emergency indicators that are critical for early detection of complications.

However, the study also identified important gaps in knowledge. Only 50% of participants correctly answered questions related to the side effects of spinal anesthesia and morphine. This indicates insufficient understanding of pharmacological and anesthetic complications, which may place patients at risk if adverse effects are not recognized early. Similarly, knowledge related to pain assessment using verbal scales and early breastfeeding initiation was

lower compared to other items. These findings may be attributed to inadequate specialized training and lack of updated clinical guidelines.

Concerning clinical practice, the findings showed that participants performed well in immediate postoperative procedures such as checking vital signs upon patient arrival from the operating room (93.3%) and assessing the incision site (90%). This reflects acceptable adherence to routine postoperative nursing care standards. Monitoring vaginal bleeding, level of consciousness, and urine output also demonstrated relatively good practice levels among participants.

Despite these positive findings, some important nursing interventions were poorly practiced. Only 53.3% of participants encouraged leg movement for prevention of deep vein thrombosis (DVT), and only 50% provided health education regarding danger signs before discharge. These findings are concerning because patient education and early mobilization are considered essential components of quality post-cesarean care. Poor performance in these areas may result from work overload, shortage of staff, or inadequate emphasis on patient-centered care during training programs.

The study further demonstrated a statistically significant relationship between training and level of clinical practice ($\chi^2 = 7.42$, $p = 0.006$). Participants who had received previous training showed significantly better practice compared to untrained staff. This finding highlights the critical role of continuous professional training in improving nursing performance and quality of care. It also suggests that implementing regular educational programs and workshops could enhance clinical competence among nurses and midwives working in postnatal wards.

In addition, the study identified a strong positive correlation between knowledge and practice scores ($r = 0.68$, $p = 0.001$). This indicates that higher

knowledge levels were associated with better clinical practice. The finding supports the assumption that improving theoretical understanding contributes directly to improving nursing performance and patient outcomes. Therefore, strengthening educational preparation and evidence-based training may positively influence the quality of immediate post-cesarean care.

Overall, the study findings indicate that nurses and midwives possess moderate knowledge and practice regarding immediate post-cesarean care. Nevertheless, significant deficiencies remain in specialized knowledge areas, patient education, and preventive nursing interventions. Addressing these gaps through continuous training, supervision, and implementation of standardized clinical protocols may contribute to improving maternal care and reducing postoperative complications among cesarean section patients.

4.2 Conclusion

The study concluded that both nurses and midwives at Sennar Maternity Teaching Hospital have a generally Moderate level of knowledge according to Knowledge scale (67%) regarding immediate post-cesarean section care. Although some participants demonstrated adequate understanding and appropriate practices in areas such as monitoring vital signs, wound care, pain management, infection prevention, and early ambulation, gaps were still identified in several aspects of postoperative care. These deficiencies may affect the quality of maternal care and increase the risk of complications after cesarean section.

4.3 Recommendations

1. Conduct continuous educational and training programs for nurses and midwives regarding immediate post–cesarean section care to improve their knowledge and clinical practices.
2. Develop and implement standardized protocols and guidelines for immediate post–cesarean section care in all maternity units and hospitals.
3. Provide regular supervision and evaluation of nurses’ and midwives’ practices to ensure adherence to proper postoperative care standards.
4. Encourage healthcare institutions to organize workshops, seminars, and refresher courses focusing on infection prevention, pain management, wound care, and early ambulation after cesarean section.
5. Ensure the availability of adequate medical equipment, supplies, and educational materials necessary for effective post–cesarean care.
6. Strengthen collaboration between nurses, midwives, and obstetric healthcare teams to improve the quality of maternal care and patient outcomes.
7. Increase awareness among mothers about the importance of immediate post–cesarean care through health education and counseling programs.

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Questionnaire

University of Sinnar

Faculty of Nursing Science

Assessment of knowledge and Practice of Nurses and Midwives Regarding Immediate Post Cesarean Care At Obstetrical and Gynecological Sinnar Teaching Hospital

Section 1: General Information

1. Age: _____

2. Profession:

a. Nurse ☐

b. Midwife ☐

3. Educational Qualification:

a. Diploma ☐

b. Bachelor's ☐

c. Master's ☐

4. Years of Experience:

a. Less than 1 year ☐

b. 1–3 years ☐

c. 4–6 years ☐

d. More than 6years ☐

5. Workplace:

a. Postnatal Ward ☐

b. Recovery Room ☐

c. Operating Theatre ☐

d. Delivery Room ☐

6. Have you received training on post-cesarean care?

a. Yes ☐

b. No ☐

Section 2: Knowledge

7. Vital signs should be monitored immediately after surgery every:

a. 15 minutes ☐

b. 30 minutes ☐

c. 60 minutes ☐

8. Pain should be assessed after delivery using:

a. Verbal pain scale ☐

b. Observation only ☐

c. Family report ☐

9. Morphine and spinal anesthesia may cause:

a. Respiratory depression ☐

b. Hypertension ☐

c. Diarrhea ☐

10. The most important indicator of internal bleeding is:

a. Increased pulse rate ☐

b. Fever ☐

c. Headache ☐

11. The first sign of wound infection is:

- a. Redness and swelling ☐
- b. Headache ☐
- c. Loss of appetite ☐

12. When should the surgical dressing be checked for bleeding?

- a. Immediately and repeatedly ☐
- b. Hypertension ☐
- c. Diarrhea ☐

13. Signs of thrombosis:

- a. Pain and swelling in one leg ☐
- b. Itching ☐
- c. Vomiting ☐

14. To reduce the risk of leg thrombosis:

- a. Early mobilization ☐
- b. Increased fluids ☐
- c. Fasting ☐

15. When is uterine massage recommended?

- a. Soft uterus ☐
- b. Firm uterus ☐
- c. Painful uterus ☐

16. If the uterus is soft, the first action is:

- a. Changing the dressing ☐
- b. Massaging the uterus ☐
- c. Giving painkillers ☐

17. Barriers to improving post-cesarean care:

- a. Work pressure ☐
- b. Lack of supplies and equipment ☐
- c. Lack of medical staff ☐
- d. Lack of clear protocols ☐

Section 3: Practice

18. Hands are washed before and after every procedure:

- a. Always ☐
- b. Never ☐
- c. Sometimes ☐

19. The uterus is palpated to check its firmness and position:

- a. Always ☐
- b. Never ☐
- c. Sometimes ☐

20. It is checked that intravenous fluids are running according to the treatment plan:

- a. Always ☐
- b. Never ☐
- c. Sometimes ☐

21. Do you monitor vital signs during the first two hours after cesarean section?

- a. Always ☐
- b. Never ☐
- c. Sometimes ☐

**Chick list about Assessment of Practice of nurses and midwives regarding
immediate post cesarean section care.**

Procedures	Done	Not Done
1. Check vital signs immediately upon arrival from the OR.		
2. Monitor pulse, blood pressure, and respiration every 15 minutes for the first hour.		
3. Check vaginal bleeding every 15-30 minutes for the first 2 hours.		
4. Palpate the uterine fundus to determine level and consistency every 15 minutes for the first hour.		
5. Check the incision site for bleeding.		
6. Assess level of consciousness and response.		
7. Encourage deep breathing and leg movement to prevent thrombosis.		
8. Assess pain and administer analgesia as planned.		
9. Assist with early initiation of breast feeding.		
10. Monitor urine output.		
11. Document all observations and interventions in the patient's file.		
12. Educate the mother on danger signs before discharge.		